

Chap 2. Random Variable.

26.06.30 (Tue)

2.1 Def. of R.V.

Actually, r.v. is a deterministic function, that maps outcomes from a random process to real numbers.

Let $(\Omega, \mathcal{F}, P)$ be a prob. sp. where:

Ω : sample space.

$\mathcal{F}$: σ -algebra of events.

P : prob. measure.

A random variable X is a measurable ftn. that maps the sample sp. to real numbers: $X: \Omega \rightarrow \mathbb{R}$.

For X to be a measurable ftn, the inverse image of any Borel set $B \subseteq \mathbb{R}$ must be an event in $\mathcal{F}$. That is:

$$\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

Range of X :

- Countably infinite $\rightarrow X$ is discrete.
- interval $\rightarrow$ continuous.
- mixed. (e.g., $\{1, 2, (5, 10)\}$).

2.2 Prob. density ftn, " distribution ftn.

Discrete $\rightarrow$ pdf. $f(x)$ should satisfy:

(1) $\forall x \in \mathbb{R}, f(x) \geq 0.$

(2) For all possible value $x_1, x_2, \dots$ of r.v X ,

$$f(x_i) \geq 0 \text{ and } \sum_{x_i} f(x_i) = 1.$$

②. Discrete r.v. $X \rightarrow$ pdf? pmf?
 $\hookrightarrow f(x) = P(X=x) = f(x).$

Conti. r.v. $X \rightarrow$ give the prob. to the range.

$$\Rightarrow (1) \forall x \in \mathbb{R}, f(x) \geq 0$$

$$(2) \int_{-\infty}^{\infty} f(x) dx = 1.$$

$$\text{For } -\infty < a \leq b < \infty, \int_a^b f(x) dx = P(a \leq X \leq b).$$

$$\text{Concisely, } P(X \in A) = \begin{cases} \sum_{x_i \in A} f(x_i), & \text{if } X \text{ is discrete.} \\ \int_A f(x) dx, & \text{if } X \text{ is continuous.} \end{cases}$$

Def 2.2 (Cumulative distribution ftn.)

$$F(x) := P(X \leq x)$$

Thm 2.1 $F(x)$ is a cum. dist. ftn. of r.v. X iff

$$\lim_{x \rightarrow -\infty} F(x) = 0, \lim_{x \rightarrow \infty} F(x) = 1, \lim_{h \rightarrow 0^+} F(x+h) = F(x), F(a) \leq F(b) \text{ if } a \leq b.$$

$$\text{So, } F(x) = \begin{cases} \sum_{x_i \leq x} f(x_i), & X: \text{discrete} \\ \int_{-\infty}^x f(t) dt, & X: \text{continuous.} \end{cases}$$

$$\Rightarrow P(a \leq X \leq b) = F(b) - F(a).$$

Thm. 2.2

If $f(x)$ and $F(x)$ are pdf. and cdf. of r.v. X ,
 then $f(x) = \frac{d}{dx} F(x)$.

(pf) $F(x) = \int_{-\infty}^x f(t) dt$. Thus, $f(x) = \frac{d}{dx} F(x)$.

ex. 2.4

$$F(x) = 1 - e^{-\lambda x} \quad (x > 0, \lambda > 0; \text{const.})$$

$$\Rightarrow f(x) = \frac{d}{dx} F(x) = \frac{1}{\lambda} e^{-\lambda x}, \quad x > 0.$$

2.3 ^(2.3.1) Joint dist, Joint. Conditional prob. dist.

R.V. vector $X = (X_1, X_2, \dots, X_k)$

Def 2.3 (Joint Prob. density ftn.)

For two r.v. X and Y , the joint-pdf $f_{X,Y}(x,y)$ is

$$f_{X,Y}(x,y) = P(X=x, Y=y) \text{ if } X, Y \text{ are discrete.}$$

$$f_{X,Y}(x,y) \text{ s.t. } P[(X,Y) \in A] = \iint_A f_{X,Y}(x,y) dx dy.$$

if X, Y are continuous.

and... for $\forall x, y \in \mathbb{R}$,

$$(1) f_{X,Y}(x,y) \geq 0$$

$$(2) \left\{ \begin{array}{l} \sum_{\forall x,y} f_{X,Y}(x,y) = 1 \quad (\text{Discrete}) \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1 \quad (\text{Continuous}). \end{array} \right. \text{ hold.}$$

④. Generally,
 $f(x_1, x_2, \dots, x_k)$ satisfy:

(1) $\forall (x_1, x_2, \dots, x_k) \in \mathbb{R}^k, f(x_1, x_2, \dots, x_k) \geq 0.$

(2)
$$\begin{cases} \sum_{\forall (x_1, x_2, \dots, x_k)} f(x_1, x_2, \dots, x_k) = 1 & \text{if discrete.} \\ \int \dots \int_{\mathbb{R}^k} f(x_1, x_2, \dots, x_k) dx_1 dx_2 \dots dx_k = 1 & \text{if conti.} \end{cases}$$

Def. 2-4.

Joint pdf (jpdf) of r.v.s $X_1, X_2, \dots, X_k$ is:

$$F(x_1, x_2, \dots, x_k) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_k \leq x_k)$$

$$\Rightarrow F(x_1, x_2, \dots, x_k) = \begin{cases} \sum_{\forall t_i \leq x_i} f(t_1, t_2, \dots, t_k) & \text{(discrete).} \\ \int_{-\infty}^{x_k} \dots \int_{-\infty}^{x_1} f(t_1, t_2, \dots, t_k) dt_1 \dots dt_k & \text{(conti.)} \end{cases}$$

If r.v. vector is conti., then

$$f(x_1, \dots, x_k) = \frac{\partial^k}{\partial x_k \dots \partial x_1} F(x_1, x_2, \dots, x_k) \quad \text{holds.}$$

Thm 2.3 for two variable r.v. (X_1, X_2) ,

$F(x_1, x_2)$ is a jpdf of (X_1, X_2) iff

$$\forall x_1, \lim_{x_2 \rightarrow -\infty} F(x_1, x_2) = F(x_1, -\infty) = 0$$

$$\forall x_2, \lim_{x_1 \rightarrow -\infty} F(x_1, x_2) = F(-\infty, x_2) = 0.$$

$$\lim_{x_1 \rightarrow +\infty, x_2 \rightarrow +\infty} F(x_1, x_2) = F(\infty, \infty) = 1.$$

$$\forall a < b, c < d; F(b, d) - F(b, c) - F(a, d) + F(a, c) \geq 0.$$

$$\forall x_1, x_2; \lim_{h \rightarrow 0^+} F(x_1, x_2 + h) = \lim_{h \rightarrow 0^+} F(x_1 + h, x_2) = F(x_1, x_2).$$

2-3-2 marginal p.d.

$$f_{X,Y}(x,y) \rightsquigarrow f_X(x) \quad ; \text{ marginal p.d.f.} \\ \rightsquigarrow f_Y(y).$$

Thm. 2.4

(i) Discrete Case

$$f_X(x) = \sum_y f_{X,Y}(x,y), \quad f_Y(y) = \sum_x f_{X,Y}(x,y)$$

(ii) Continuous Case

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

(pf.) $f_X(x) = \frac{d}{dx} F_X(x)$

$$= \frac{d}{dx} P(X \leq x, -\infty < Y < \infty)$$

$$= \frac{d}{dx} \left[\int_{-\infty}^x \int_{-\infty}^{\infty} f_{X,Y}(u,y) dy du \right]$$

$$= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \quad \square$$

For jpdf $f(x_1, x_2, \dots, x_k)$, marginal pdf of X_i ($1 \leq i \leq k$) is

(i) discrete. $f_{X_i}(x_i) = \sum_{x_1} \dots \sum_{x_{i-1}} \sum_{x_{i+1}} \dots \sum_{x_k} f(x_1, x_2, \dots, x_k)$

(ii) conti.

$$f_{X_i}(x_i) = \int \dots \int f(x_1, \dots, x_k) dx_1 \dots dx_{i-1} dx_{i+1} \dots dx_k.$$

6.

(2.3.3) Conditional p.d.f.

$Y|X=x$, given. (sometimes, " $Y|x$ ").

Def. 2.5 $Y|x \Rightarrow$ conditional pdf is

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Generally, $k > 2$, jpmf: $f(x_1, \dots, x_k); X_1=x_1, X_2=x_2, \dots, X_k=x_k. (i < k).$

$$f(x_{i+1}, \dots, x_k | x_1, x_2, \dots, x_i) = \frac{f(x_1, x_2, \dots, x_k)}{f(x_1, x_2, \dots, x_i)}$$

where

(i) $f(x_1, \dots, x_i) = \sum_{x_{i+1}} \dots \sum_{x_k} f(x_1, x_2, \dots, x_k)$
discrete

(ii) $f(x_1, \dots, x_i) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_k) dx_{i+1} \dots dx_k$
Conti.

(2.3.4) Independent R.V.

Def. 2.6 (Independent).

$$P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B).$$

Thm. 2.5. $X \perp Y$ iff. $f_{X,Y}(x,y) = f_X(x) f_Y(y)$

(pf). $\int_B \int_A f_{X,Y}(x,y) dx dy = P(X \in A, Y \in B)$

$$= P(X \in A) \cdot P(Y \in B)$$

$$= \int_A f_X(x) dx \cdot \int_B f_Y(y) dy$$

$$= \iint_{BA} f_X(x) \cdot f_Y(y) dx dy \text{ holds iff}$$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y).$$

(7)

Note that $f_{X,Y}(x,y) = f_X(x) f_Y(y)$.

$$\text{iff } F_{X,Y}(x,y) = F_X(x) F_Y(y)$$

So r.v.s. $X_1, X_2, \dots, X_k$ are independent. iff.

$$f(x_1, x_2, \dots, x_k) = f_{X_1}(x_1) f_{X_2}(x_2) \dots f_{X_k}(x_k).$$

Thus, if we get random samples (size: n) $X_1, X_2, \dots, X_n$ from the population which has a pdf $f(x)$, the jpdf is

$$f(x_1, x_2, \dots, x_n) = f(x_1) f(x_2) \dots f(x_n).$$

2.4 Expectation.

$$\text{Def. 2.7} \quad E(X) = \begin{cases} \sum_{x_i} x_i f(x_i) & (\text{Discrete}) \\ \int_{-\infty}^{\infty} x f(x) dx & (\text{Continuous}). \end{cases}$$

and $E(|X|) < \infty$ must be.

If $E(X)$ is not a finite num, then we say "expectation $\neq$ "

$$\Rightarrow E_X[g(X)] = E_Y(Y) = \begin{cases} \sum_{y_i} y_i f_Y(y_i) \\ \int_{-\infty}^{\infty} y f_Y(y) dy. \end{cases}$$

($Y = g(X)$).

$$\Rightarrow E[g(X)] = \begin{cases} \sum_{x_i} g(x_i) f_X(x_i) \\ \int_{-\infty}^{\infty} g(x) f_X(x) dx. \end{cases}$$

8.

Thm 2.6. For constants a, b, c ,

$$E(c) = c, \quad E(ax+b) = aE(X) + b.$$

(pf). $E(c) = \int_{-\infty}^{\infty} cf(x)dx = c, \quad E(ax+b) = \int_{-\infty}^{\infty} (ax+b)f(x)dx$

$$= a \int_{-\infty}^{\infty} xf(x)dx + b \int_{-\infty}^{\infty} f(x)dx$$

$$= aE(X) + b.$$

(i) $E[g(X_1, \dots, X_n)] = \sum_{x_1} \dots \sum_{x_n} g(x_1, x_2, \dots, x_n) f(x_1, x_2, \dots, x_n).$

(i') $E[g(X_1, \dots, X_n)] = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} g(x_1, x_2, \dots, x_n) f(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n.$

(1) $E(ax+bY) = aE(X) + bE(Y). \quad (a, b: \text{const.})$

(2) $E(a_1X_1 + \dots + a_nX_n) = a_1E(X_1) + \dots + a_nE(X_n). \quad (a_n: \text{const.})$

Thm. 2.7 If $X \perp Y$, then

$$E(XY) = E(X)E(Y),$$

If $g(X) \perp h(Y)$, then

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)].$$

(pf)

$$E(XY) = \sum_{x,y} xy f_{X,Y}(x,y)$$

$$= \sum_{x,y} xy f_X(x) f_Y(y)$$

$$= E(X)E(Y).$$

$$E[g(X)h(Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y) f_{X,Y}(x,y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y) f_X(x) f_Y(y) dx dy$$

$$= E[g(X)]E[h(Y)].$$

* inverse does not hold.

2.4.2 Var, Cov.

Def. 2.8 $\text{Var}(X) = E[(X - E[X])^2]$

$$\sigma_x = \sqrt{\text{Var}(X)}.$$

Standardization. $\rightarrow Z = \frac{X - E[X]}{\sigma_x}$
(Z-transform)

$$\begin{aligned}\text{Var}(X) &= E[(X - E[X])^2] = E[X^2 - 2E[X]X + (E[X])^2] \\ &= E[X^2] - (E[X])^2.\end{aligned}$$

Thm. 2.8

(1) If $Y = aX + b$, then $\text{Var}(Y) = a^2 \text{Var}(X)$.

(2) If $X_1 \perp X_2 \perp \dots \perp X_n$, then $\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i)$.

(pf) (1) trivial. (2) Use mathematical induction.

Def 2.9 (Covariance)

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y].\end{aligned}$$

$\rightarrow \text{Cov}(X, X) = \text{Var}(X)$, If $X \perp Y$, then $\text{Cov}(X, Y) = 0$. ($\nLeftarrow$).

10.

Thm. 2.9.

$$\text{Cov}(aX+b, cY+d) = ac \text{Cov}(X, Y)$$

(pf). trivial.

Thm 2.10.

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{j < k} \text{Cov}(X_j, X_k).$$

$$(pf). \text{Var}\left(\sum_{i=1}^n X_i\right) = E\left[\left(\sum_{i=1}^n X_i\right)^2\right] - \left[E\left[\sum_{i=1}^n X_i\right]\right]^2$$

$$= E\left[\sum_{i=1}^n X_i^2 + 2 \sum_{j < k} X_j X_k\right] - \left[\sum_{i=1}^n (E[X_i])^2 + 2 \sum_{j < k} E[X_j] E[X_k]\right]$$

$$= \sum_{i=1}^n [E[X_i^2] - (E[X_i])^2] + 2 \sum_{j < k} [E(X_j X_k) - E(X_j) E(X_k)]$$

$$= \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{j < k} \text{Cov}(X_j, X_k). \quad \square$$

2.4.3 Conditional Expectation.

Def. 2.10 (Conditional Expectation).

$$E[Y|X=x] = \begin{cases} \sum y_j f_{Y|X}(y_j|x) & X, Y : \text{Discrete.} \\ \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy & X, Y = \text{Conti.} \end{cases}$$

Thm 2.11 (Double expectation).

$$E[E[Y|X]] = E[Y].$$

(pf). (Assume that $E[|Y|] < \infty$.)

$$\begin{aligned} E[E[Y|X]] &= \int_{-\infty}^{\infty} E[Y|x] f_X(x) dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{Y|X}(y|x) f_X(x) dy dx. \\ &= \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f(x,y) dx dy \quad (\because E[Y] \text{ exist}) \\ &= \int_{-\infty}^{\infty} y f_Y(y) dy = E[Y]. \end{aligned}$$

Generally, $E[g(X,Y)] = E_X[E[g(X,Y)|X]]$.

Ex 2.24 $f_{X,Y} = \begin{cases} \frac{1}{2} & , 0 \leq x \leq y \leq 2, \\ 0 & , \text{o.w.} \end{cases}$

$$f_X(x) = \int_x^2 \frac{1}{2} dy = \frac{1}{2}(2-x), \quad 0 \leq x \leq 2,$$

$$f_Y(y) = \int_0^y \frac{1}{2} dx = \frac{1}{2}y, \quad 0 \leq y \leq 2.$$

$$\Rightarrow f_{Y|X}(y|x) = \frac{1/2}{(2-x)/2} = \frac{1}{2-x}, \quad x \leq y \leq 2.$$

$$\leadsto E[Y|X=x] = \frac{2+x}{2} \quad \text{where } 0 \leq x \leq 2.$$

$$E[E[Y|X=x]] = \int_0^2 \frac{1}{2}(2+x) \cdot \frac{1}{2}(2-x) dx = \frac{4}{3} = E[Y].$$

⑫.

Ex. 2.25

$$f_X(x) = \frac{1}{2} e^{-\frac{x}{2}}$$

$$f_{Y|X}(y|x) = \begin{cases} \frac{1}{x}, & 0 \leq y \leq x \\ 0, & \text{o.w.} \end{cases} \Rightarrow E[Y] = ?$$

$$E[Y|X=x] = \int_0^x y \cdot \frac{1}{x} dy = \frac{1}{2} x, \quad x > 0.$$

$$E[Y] = E[E[Y|X]] = \int_0^\infty \frac{1}{2} x \cdot \frac{1}{2} e^{-\frac{x}{2}} dx = \int_0^\infty z e^{-z} dz = 1.$$

Thm. 2.12

$$X \perp\!\!\!\perp Y \Rightarrow E[Y|x] = E[Y], \quad E[X|y] = E[X].$$

(pf).

$$\begin{aligned} E[Y|x] &= \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy \\ &= \int_{-\infty}^{\infty} y \frac{f_X(x) f_Y(y)}{f_X(x)} dy = \int_{-\infty}^{\infty} y f_Y(y) dy = E[Y]. \quad \square \end{aligned}$$

Def. 2.11 (Conditional Variance)

$$\text{Var}(Y|x) = E[(Y - E[Y|x])^2 | x].$$

Note that

$$\begin{aligned} \text{Var}(Y|x) &= E[(Y - E[Y|x])^2 | x] \\ &= E[Y^2|x] - (E[Y|x])^2. \end{aligned}$$

Ex. 2.26.

$$f_{Y|X}(y|x) = xe^{-xy} \rightarrow E[Y|x] = \frac{1}{x}. \quad (x > 0, y > 0).$$

$$\begin{aligned} E[Y^2|x] &= \int_{-\infty}^{\infty} y^2 f_{Y|X}(y|x) dy \\ &= \int_0^{\infty} xy^2 e^{-xy} dy = -y^2 e^{-xy} \Big|_0^{\infty} + \int_0^{\infty} 2y e^{-xy} dy \\ &= \frac{2}{x^2}. \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Var}(Y|x) &= E[Y^2|x] - (E[Y|x])^2 \\ &= \frac{2}{x^2} - \left(\frac{1}{x}\right)^2 = \frac{1}{x^2}. \end{aligned}$$

Decomposition of conditional variance.

Thm 2.13

$$\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X])$$

$$\begin{aligned} \text{(pf). } E[\text{Var}(Y|X)] &= E[E[Y^2|X] - (E[Y|X])^2] \\ &= E[Y^2] - (E[E[Y|X]^2]) \\ &= E[Y^2] - (E[Y])^2 - (E[E[Y|X]]^2 - E[E[Y|X]]^2) \\ &= \text{Var}(Y) - \text{Var}(E[Y|X]). \end{aligned}$$

 $\leadsto$ Note that

$$E[\text{Var}(Y|X)] \leq \text{Var}(Y).$$

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2.4.4 Probability inequality.

Thm. 2.14 (Markov Inequality)

For a real-valued fcn $u(x) \geq 0$ and constant $c > 0$,

$$P[u(X) \geq c] \leq \frac{E[u(X)]}{c}.$$

(pf). $A := \{x \mid u(x) \geq c\}$.

$$E[u(X)] = \int_{-\infty}^{\infty} u(x) f(x) dx$$

$$= \int_A u(x) f(x) dx + \int_{A^c} u(x) f(x) dx$$

$$\geq \int_A u(x) f(x) dx \geq \int_A c f(x) dx = c P[X \in A]$$

$$= c P[u(X) \geq c]. \quad \square$$

If we take $u(x) = |x|^r$,

$$P[|X| \geq c] \leq \frac{E[|X|^r]}{c^r}.$$

Thm. 2.15 (Chebyshev)

$X: \mu, \sigma^2 < \infty$, for any $k > 0$,

$$P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}$$

(pf). Take $u(x) = (x - \mu)^2$, $c = k^2 \sigma^2$.

$$\Rightarrow P[(X - \mu)^2 \geq k^2 \sigma^2] \leq \frac{E[(X - \mu)^2]}{k^2 \sigma^2} = \frac{1}{k^2}.$$